Scenario-Based Uncertain Topology and Extended Logic Valuation

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Abstract

This paper extends Scenario-Based Uncertain Set Theory and Extended Logic into general topology. Rather than inflating the framework with ad-hoc fuzzy membership functions or complex parameterized soft axioms, we construct Uncertain Topology directly from First Principles as a topological structure over an Uncertain Universe X. By evaluating structural topological statements through the Extended Logic truth valuation operator $\mathcal{V}(P)$, we establish that the assertion "$\mathcal{T}$ is a Topology" holds with a scenario measure $\mathcal{V} \in [0, 1]$. We prove the Topological Collapse Theorem: when the uncertain cardinalities of all open sets collapse into deterministic natural numbers ($\text{ucard}(A) \in \mathbb{N}$), the uncertain topology degenerates canonical-wise into a classical static topology. Finally, we formulate a

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lightweight, computationally efficient metric—the Topological Uncertainty Index $\mu_U(\mathcal{T})$—which quantifies structural variation directly on cardinality domains without incurring computational noise or density estimation overhead.

Keywords: Uncertain Topology, Extended Logic, Truth Valuation, Configuration Space, Topological Collapse, Uncertain Cardinality, Scenario Measure, First Principles.

Chapter 1

Extended Logic and Uncertain Sets Review

Definition 1.1 (Scenario Space and Truth Valuation)**.** *Let* $(\Omega, \mathcal{F}, \lambda)$ *be a measure space representing the scenario domain. For any structural proposition* P*, its scenario truth function is* $v(P, \cdot) : \Omega \to \{0, 1\}$*. The global truth valuation operator* $\mathcal{V}$ *maps* P *to* $[0, 1]$ *via:*

$$\mathcal{V}(P) := \frac{\lambda(\{\omega \in \Omega : v(P, \omega) = 1\})}{\lambda(\Omega)}$$

Definition 1.2 (Uncertain Set and Configuration Space)**.** *Let* U *be a non-empty universal set. An uncertain set* $S = \{x_i : i \in I\}$ *on* U *induces a family of classical set realizations across scenarios* $\omega \in \Omega$*, forming its Configuration Space:*

$$\text{Conf}(S) = \{S^*(\omega) \subseteq U : \omega \in \Omega\}$$

The uncertain cardinality $\text{ucard}(S)$ *is an uncertain natural number whose possibility domain is:*

$$\text{Val}(\text{ucard}(S)) = \{\text{card}(S^*(\omega)) : S^*(\omega) \in \text{Conf}(S)\} \subseteq \mathbb{N}$$

Chapter 2

Uncertain Topology via Extended Logic

Definition 2.1 (Uncertain Open Sets and Structural Topology)**.** *Let X be an uncertain universal set. A collection $\mathcal{T}$ of uncertain open subsets of X is called an* Uncertain Topology *on X. For each scenario $\omega \in \Omega$, the realized collection of classical sets is denoted by $\mathcal{T}^*(\omega) \in \mathrm{Conf}(\mathcal{T})$.*

Proposition 2.1 (Extended Logic Valuation of Topological Axioms)**.** *Let $\Phi(\mathcal{T})$ be the proposition asserting that $\mathcal{T}$ satisfies classical topology axioms:*

i) $\emptyset \in \mathcal{T}^*(\omega)$ and $X^*(\omega) \in \mathcal{T}^*(\omega)$.

ii) Arbitrary unions of sets in $\mathcal{T}^*(\omega)$ remain in $\mathcal{T}^*(\omega)$.

iii) Finite intersections of sets in $\mathcal{T}^*(\omega)$ remain in $\mathcal{T}^*(\omega)$.

The structural truth degree of $\mathcal{T}$ being a topology is evaluated under Extended Logic as:

$$\mathcal{V}(\mathcal{T} \text{ is a Topology}) = \frac{\lambda(\{\omega \in \Omega : \Phi(\mathcal{T}^*(\omega)) = 1\})}{\lambda(\Omega)} \in [0, 1]$$

If $\mathcal{V}(\mathcal{T} \text{ is a Topology}) = 1$, $\mathcal{T}$ is said to be an Absolutely Consistent Uncertain Topology.

Theorem 2.1 (Topological Collapse Theorem)**.** *Let $\mathcal{T}$ be an uncertain topology on X. If for every uncertain open set $A \in \mathcal{T}$, its uncertain cardinality possesses a singleton value domain:*

$$\mathrm{Val}(\mathrm{ucard}(A)) = \{k_A\}, \quad k_A \in \mathbb{N}$$

then structural fluctuation across scenario space Ω vanishes completely, and $\mathcal{T}$ collapses canonically into a classical static topology.

Proof. For each $A \in \mathcal{T}$, $\mathrm{Val}(\mathrm{ucard}(A)) = \{k_A\}$ implies that $\mathrm{card}(A^*(\omega)) = k_A$ for all scenarios $\omega \in \Omega$. Since no element realization induces cardinality expansion or state collapse, each configuration space $\mathrm{Conf}(A)$ reduces to a unique deterministic set A^*. Consequently, $\mathrm{Conf}(\mathcal{T})$ contains exactly one realized topology $\mathcal{T}^*$, proving that $\mathcal{T}$ collapses into a classical static topology. □

Chapter 3

Topological Uncertainty Index

Definition 3.1 (Topological Uncertainty Index)**.** *Let $\mathcal{T}$ be an uncertain topology on X. The* Topological Uncertainty Index $\mu_U(\mathcal{T}) \in [0, 1)$ *quantifies structural uncertainty directly over the cardinality domains* $\mathrm{Val}(\mathrm{ucard}(A))$ *of constituent open sets $A \in \mathcal{T}$:*

$$\mu_U(\mathcal{T}) = 1 - \frac{\sum_{A \in \mathcal{T}} \min \mathrm{Val}(\mathrm{ucard}(A))}{\sum_{A \in \mathcal{T}} \sum_{k \in \mathrm{Val}(\mathrm{ucard}(A))} k}$$

Proposition 3.1 (Properties of $\mu_U(\mathcal{T})$)**.** *The metric $\mu_U(\mathcal{T})$ satisfies:*

a) $\mu_U(\mathcal{T}) = 0$ if and only if $\mathcal{T}$ collapses into a classical static topology.

b) $\mu_U(\mathcal{T}) \to 1$ as cardinality variability across scenarios expands infinitely.

c) Computation of $\mu_U(\mathcal{T})$ retains $\mathcal{O}(1)$ algorithmic complexity via Lazy Evaluation on index domains.

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